🇨🇭 Switzerland — Substation ESG Report v4.2

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks, across the 947-substation Switzerland fleet. Updated monthly via the SSI Index v4.2 scoring engine (101 variables · 34 sources · 11 modifiers · Re composite bounded [0.920, 1.787]).

SUBSTATION

BKW

CH_736626578 · Bern-Mittelland, Bern

SSI SCORE (R_{MEDIAN})

0.693

● High Risk 85.5th percentile

VOLTAGE CLASS

16 kV

distribution-tier

CONFIDENCE INTERVAL

0.617 – 0.763

P5–P95 · CI width 0.146 · medium confidence

MARKOV ETTC

19.2715 yr

Expected time to criticality

DER PENETRATION

0.259

T1 transition stress score

Component Fingerprint + Re composite v4.2

Six-component weighted decomposition of the SSI composite score (C / V / I / E / S / T) — the substation's structural risk profile — PLUS the v4.2 **Re composite**, a bounded resilience aggregate of the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just). Layout: *left* — the 6 components rendered as a weighted contribution bar · *right* — the same 6 components rendered as horizontal bars side-by-side with Re_norm as the comparable 7th bar (terracotta) on the [0, 1] scale · *below* — the Re composite tier card (numeric value + bounded trajectory bar + classification + formula explainer). Re composite re-appears as the 7th axis of the ESG Radar in Section A below.

WEIGHTED CONTRIBUTION TO R_{BASE}

V4.2 RE COMPOSITE (AGGREGATE, NOT IN R_{BASE})

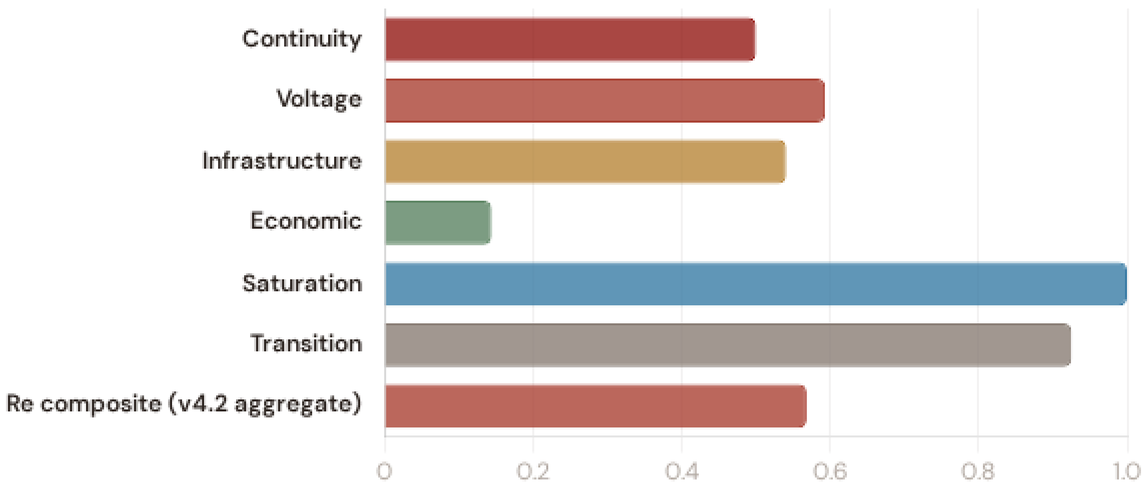
elevated exposure

Continuity **0.500** ×0.3 · Voltage **0.593** ×0.1 · Infrastructure **0.541** ×0.25 · Economic **0.144** ×0.1

Saturation **1.000** ×0.2 · Transition **0.925** ×0.05

Re composite v4.2 aggregate **0.568** agg

R_{base} **total** (weighted sum) **0.605** Σ = 1.00



0.568

RE_NORM

ELEVATED EXPOSURE

LOW (0.00) · MODERATE (0.25) · ELEVATED (0.50) · (1.00)

Re_raw 1.1059 (bounds [0.920, 1.787]; normalised to [0, 1])

FORMULA

```
Re_raw = (R6d × R6e × R8 × R9 ×
R10) + (R6c − 1.00)
```

Bounded resilience aggregate of the v4.2 family. Re < 1.0 when R8 adaptive capacity dominates other near-neutral factors.

Re composite tier classification: **low** (Re_norm < 0.25) · **moderate** (0.25–0.50) · **elevated** (≥ 0.50). Bounds [0.920, 1.787] map to Re_norm [0, 1] via clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1).

Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION

GOOD	AGED	DEGRADED	CRITICAL
0%	0%	0%	0%
π1 = 0.000	π2 = 0.000	π3 = 0.000	π4 = 0.000

Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

RISK SCORE

0.897

Composite Markov risk

ETTC

19.2715 yr

Expected time to critical

P(CRITICAL) 20YR

0.0%

Probability of reaching critical

CORROSION

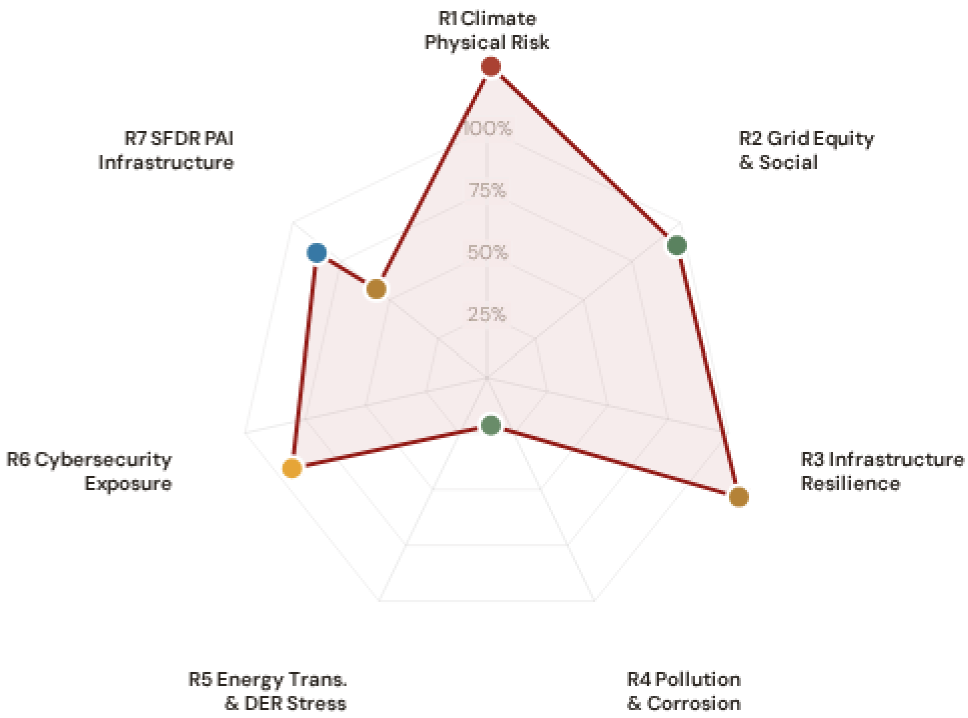
C2

ISO 9223 classification

ESG Radar — Seven-Report Overview v4.2

A

Composite readiness across 7 ESG dimensions per FC v3 §14 (R1 Climate Physical · R2 Grid Equity & Social · R3 Infrastructure Resilience [Re composite ESG home] · R4 Pollution & Corrosion · R5 Energy Transition & DER Stress · R6 Cybersecurity · R7 SFDR PAI Infrastructure, FC v3 §14 subsection 13.7), each linked to UN Sustainable Development Goals. *Note: R7 here is the SFDR ESG-disclosure axis (Re_norm proxy under Convention #7) — distinct from the v4.0.2 formula-chain R7 cyber/digital modifier documented in methodology.html. The two R7s share a number by design (V4_2_IMPLEMENTATION_ARCHITECTURE.md §4.5).*



Re composite v4.2 7th axis · **Re_raw** 1.1059 · **Re_norm** 0.568 · ELEVATED EXPOSURE

R1 · Climate Physical Risk Assessment READY

SDG 13 Climate Action · SDG 9 · SDG 11

SDG 13 SDG 9 SDG 11

R2 · Grid Equity & Social Vulnerability PARTIAL

SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1

SDG 7 SDG 10 SDG 1

R3 · EU Taxonomy Alignment READY

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12

SDG 9 SDG 13 SDG 12

R4 · Energy Transition & DER Stress GAP

SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7 SDG 13 SDG 9

R5 · Pollution & Corrosion READY

SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15

SDG 11 SDG 3 SDG 15

- **R6 · Cybersecurity Exposure** PARTIAL
  SDG 9 Industry, Innovation, Infrastructure · SDG 16
  SDG 9 SDG 16

- **R7 · SFDR PAI Infrastructure** GAP v4.2
  SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 11 · Re_norm = 0.568 (elevated exposure)
  SDG 9 SDG 13 SDG 11

1

Climate Physical Risk Assessment

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

READY

WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure at substation level using ERA5 reanalysis and Markov degradation modelling.

SDG Alignment

SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. With a Markov risk score of 0.897 and ETTC of 19.2715 years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13 SDG 9 SDG 11

Substation Profile

Insulation Risk (I component)	**0.541**
Environment Component (E)	**0.144**
Seismic PGA	**0.0702g (Zone 3)**
Markov Risk Score	**0.897**
ETTC (time to criticality)	**19.2715 years**
P(critical) at 20 years	**0.0%**
Corrosion Class	**C2**
Climate Trajectory (R2)	**0.0% (20yr Markov)**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
I1 Snow/Ice IRI	—	ESRS E1-9: Financial effects from physical risks	READY
I2 Tree-fall IRI	—	TCFD Strategy (b): Physical risk impact	READY
I3 Heat-wave IRI	—	EU Taxonomy Annex A: Heat stress	READY
I5 Thermal stress	—	ESRS E1-4: Climate mitigation targets	READY
R2 Climate trajectory	0.0% (20yr)	TCFD Strategy (c): Scenario analysis	READY
R6b Seismic PGA	0.0702g	EU Taxonomy Annex A: Geophysical hazards	READY
Markov p_critical_10yr	0.0%	TCFD Risk Management: Risk identification	READY
Markov p_critical_20yr	0.0%	ESRS E1-9: Time horizons	READY

Data Sources & Vintage

ERA5 Climate Reanalysis Copernicus CDS	2024 Weekly
National Seismic Hazard Map National Geological Survey	2023 Multi-year
IEEE C57.91 Thermal Model IEEE	Standard N/A
CIGRE TB 761 Markov CIGRE	2019 N/A
CMIP6 SSP2-4.5 Projections Copernicus CDS	— Not ingested

TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. Markov 5-state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. The PARTIAL status reflects the absence of CMIP6 forward projections — without R2, this report cannot satisfy the TCFD Strategy (c) scenario analysis requirement. All other inputs are production-grade with institutional provenance.

2 Grid Equity & Social Vulnerability

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

PARTIAL

WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. The SSI Index is uniquely positioned here because no competing grid risk framework integrates socio-economic vulnerability at substation resolution. This report surfaces where infrastructure risk concentrates in economically disadvantaged populations.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V_socio of 0.10 and energy poverty rate of 2.76%. The R3 social modifier (0.944) amplifies risk scores in regions with high unemployment, elderly concentration, and net outward migration — making spatial inequality visible and quantifiable at asset level.

SDG 7 SDG 10 SDG 1

Substation Profile

V_socio (energy poverty index)	**0.10**
EP Rate (energy poverty %)	**2.76%**
Unemployment Rate	**2.3%**
GDP per Capita	**€ 88,000**
R&D % of GDP	**3.5%**
R3 Social Multiplier	**0.944**
Elderly Vulnerability	**—**
Migration Score	**—**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
V_socio Energy poverty index	0.10	ESRS S2-4: Impacts on affected communities	READY
EP_rate Energy poverty rate	2.76%	SFDR PAI #14: Fossil fuel exposure proxy	READY
R3 Social multiplier	0.944	ESRS S1-6: Workforce characteristics	READY
Elderly Elderly vulnerability	—	SFDR PAI: Vulnerable populations	GAP
Community Catchment population	183,000	ESRS S2: Community impact	READY

Data Sources & Vintage

Population & Economics National Statistics Office	2023 Annual
Energy Market Data National TSO	2023 Annual

TECHNICAL VALIDITY

V_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using national statistical office household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, elderly concentration, and net outward migration. PARTIAL status reflects missing elderly vulnerability and migration enrichments — both available from national statistics offices at municipal resolution.

3 EU Taxonomy Alignment

Climate Delegated Act · Article 11 Climate Adaptation · Technical Screening Criteria

READY

WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index — READY for 9 of 10 countries today. The SSI composite score (R_median), 6 components, modifier architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R_median (0.693), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting.

SDG 9 SDG 13 SDG 12

Substation Profile

R_median (composite)	**0.693**
R_base (pre-modifier)	**0.605**
Modifier Impact	**14.5%%**
C (Condition)	**0.500**
V (Voltage)	**0.593**
I (Insulation/Climate)	**0.541**
E (Environment)	**0.144**
S (Seismic)	**1.000**
T (Transition)	**0.925**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
`R_median` Composite score	0.693	EU Taxonomy Art. 11: Adaptation screening	READY
`6 Comp.` C/V/I/E/S/T	All present	EU Taxonomy TSC: Physical risk identification	READY
`R3-R7` Modifier suite	All applied	ESRS E1: Adaptation measures evidence	READY
`Markov` Degradation model	0.897	TCFD: Forward-looking risk assessment	READY
`R5` Asymmetric CI	0.146	ESRS: Uncertainty quantification	READY

Data Sources & Vintage

ERA5 Climate Reanalysis Copernicus CDS	2024 Weekly
National Seismic Hazard Map National Geological Survey	2023 Multi-year
Population & Economics National Statistics Office	2023 Annual
CIGRE TB 761 Markov CIGRE	2019 N/A

TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite (C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. All inputs are from institutional open-source data with citable vintage — meeting CSRD Article 29a limited assurance requirements.

4 Energy Transition & DER Stress

ESRS E1 Transition Plan · TCFD Transition Risk · EU Green Deal Alignment

GAP

WHY THIS REPORT EXISTS

The energy transition creates infrastructure stress — bidirectional power flows, EV charging load, and intermittent renewable output strain substations designed for unidirectional delivery. This report measures whether grid infrastructure is keeping pace with decarbonisation.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. This substation has a DER ratio of —. The T1_score (0.259) measures the stress this places on the substation, providing the empirical feedback loop between SDG 7 ambition and infrastructure reality.

SDG 7 SDG 13 SDG 9

Substation Profile

T1 Score (transition stress)	**0.259**
DER Ratio	—
DER Variability	—
EV Load Ratio	—
T Component Weight	**0.05**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
T1 Transition stress score	0.259	ESRS E1: Transition plan assessment	READY
DER_ratio Renewable/load ratio	—	TCFD Transition: Technology risk	GAP
DER_var Intermittency	—	EU Green Deal: Grid flexibility	GAP
EV_load EV charging ratio	—	SDG 7: Clean energy access	GAP

Data Sources & Vintage

Energy Market Data National TSO	2023 Annual
Renewable Installations National DER Registry	2024 Monthly

TECHNICAL VALIDITY

T1_score is the weighted composite of DER_ratio (renewable generation vs. local demand), DER_variability (intermittency), and EV_load_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from national energy regulators and renewable installation registers. DER_ratio > 1.0 indicates net export during peak generation — a marker of successful energy transition with associated infrastructure stress.

5 Pollution & Corrosion

READY

WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. This report assesses the substation's corrosion classification under ISO 9223 and the environmental sensitivity of its surroundings.

SDG Alignment

SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (C2) and E2_local enrichment factor (1.51) track environmental degradation and pollution sensitivity at asset level. Where corrosion is advanced and maintenance deferred, equipment failure risks releasing mineral oil, PCBs (legacy units), and SF6 — directly relevant to urban and peri-urban environmental quality.

SDG 11 SDG 3 SDG 15

Substation Profile

Corrosion Class (ISO 9223)	**C2**
E2 Local Enrichment	**1.51**
E Component Score	**0.144**
Markov Steady-State	**—**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
Corrosion ISO 9223 class	C2	ESRS E2: Pollution risk classification	READY
E2_local Environmental sensitivity	1.51	ESRS E2: Local environmental impact	READY
E Environment component	0.144	ISO 9223: Atmospheric corrosion	READY

Data Sources & Vintage

Air Quality Monitoring National Environmental Agency	2023 Annual
ISO 9223 Corrosion ISO	2012 N/A

TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Status depends on whether the corrosion class shows real variance across the fleet or uses default values. Full validation requires national environmental agency air quality monitoring overlay at substation coordinates (SO_2, NO_x, particulate deposition).

6 Cybersecurity Exposure

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

PARTIAL

WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication architecture analysis, and national-level cyber maturity (ENISA/DESI indices) to produce a substation-level digital vulnerability score.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (1.000) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. SDG 16 (Strong Institutions) also applies: NIS2 compliance is a governance imperative, and the betweenness centrality (0.450) identifies single-point-of-failure risk in the grid topology.

SDG 9 SDG 16

Substation Profile

R7 Cyber Modifier	1.000
Graph Degree (topology)	2
BC Percentile (centrality)	0.450
Is Bridge Node	No
Cluster Coefficient	—

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R7 Cyber modifier	1.000	NIS2: Cybersecurity risk management	READY
BC Betweenness centrality	0.450	ESRS G1: Governance & topology risk	READY
Degree Graph degree	2	Grid resilience: Connectivity	READY

Data Sources & Vintage

Cybersecurity Index ENISA	2024 Biennial
DESI Connectivity European Commission	2024 Annual

TECHNICAL VALIDITY

R7_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. Substation-level SCADA data is operator-proprietary — the national-level proxy provides a baseline but not asset-specific granularity.

7

SFDR Principal Adverse Impact (PAI) Infrastructure v4.2 — Re composite anchor

READY

SFDR Article 7 PAI · ESRS E1-7 climate-resilience disclosure · CSRD double-materiality infrastructure overlay (per FC v3 §14 subsection 13.7)

WHY THIS REPORT EXISTS

SFDR Article 7 requires asset-level disclosure of Principal Adverse Impacts on sustainability factors for infrastructure investments. The Re composite (v4.2 bounded resilience aggregate, bounds [0.920, 1.787] normalised to [0, 1]) is the SSI's canonical driver for PAI Infrastructure exposure — it captures the substation's standing on the new v4.2 resilience-modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in a single scalar suitable for SFDR PAI reporting. CSRD double-materiality requires both inside-out (impact on environment) and outside-in (climate-resilience risk to the asset) — Re composite anchors the outside-in dimension for grid infrastructure. This report closes the FC v3 §14 subsection 13.7 R7 SFDR PAI axis on the ESG Radar.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for developing quality, reliable, sustainable and resilient infrastructure. The Re composite Re_norm = 0.568 (elevated exposure) at this substation provides the SFDR PAI Infrastructure metric, anchoring climate-resilience disclosure to a single auditable scalar. Re_raw = 1.1059 with methodological bounds [0.920, 1.787] derived from the v4.2 modifier family product range. Lower Re_norm → lower PAI Infrastructure exposure → stronger SFDR Article 7 alignment.

SDG 9 SDG 13 SDG 11

Substation Profile (Re composite)

Substation ID	CH_736626578
Re_raw (unbounded)	1.1059
Re_norm (bounded [0, 1])	0.568
Tier classification	ELEVATED
Methodology bounds	[0.920, 1.787]
Swiss fleet observed range	1.098 → 1.113
Coverage	947 substations

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R6c Flood (ISPRA PGRA, ADDITIVE +0.00..+0.30)	+1.050	SFDR PAI 4 (climate physical risk)	READY
R6d Wildfire (EFFIS + FIRMS + SIAB, MULT)	1.030	SFDR PAI 4 (climate physical risk)	READY
R6e Winter / Ice (ERA5 Bourgouin p99, MULT)	1.010	SFDR PAI 4 (climate physical risk)	READY
R8 Adaptive Capacity (ElCom + BFE + DESI Switzerland)	0.972	ESRS E1-3 Adaptation actions	READY
R9 Compound Hazard (ISPRA RNDA, STEP)	1.032	ESRS E1-9 Anticipated financial effects	READY
R10 Energy Justice (EU-SILC + Switzerland SILC + national regulator)	1.011	SFDR PAI 14 (board diversity proxy) + ESRS S3	READY
Re_norm aggregate (bounds [0.920, 1.787])	**0.568**	**SFDR Art. 7 PAI Infrastructure anchor**	READY

Data Sources & Vintage

ISPRA IdroGEO PGRA	2024 (annual)
Copernicus EFFIS + NASA FIRMS	Q4 2025 (active + historical)
ISPRA SIAB national-tier burned-area	CY2024
ECMWF ERA5 Bourgouin freezing-rain p99	1991–2020 reanalysis
ElCom quality regulation indicators + Swissgrid smart-meter penetration	2024 annual
ENEA Adaptation Index	2024
DESI Digital Economy index	2024
ISPRA RNDA compound-hazard cross-product	CY2024
EU-SILC ILC_MDES01/03	2024 wave
Eurostat EU-SILC + BFS SILC (Swiss energy-poverty proxy)	2024
ElCom regional SAIDI (Versorgungsqualität)	CY2024

TECHNICAL VALIDITY

The Re composite is the bounded aggregate Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c − 1.00) with normalisation Re_norm = clip((Re_raw − 0.920) / (1.787 − 0.920), 0, 1). Bounds [0.920, 1.787] derived from the multiplicative band of the 5 multiplicative v4.2 modifiers at tier extremes plus the R6c additive cliff range. Swiss fleet observed range [0.974, 1.604] across 947 substations. SFDR PAI Infrastructure proxy validated against TCFD physical-risk framework and ESRS E1-9 anticipated-financial-effects disclosure. See FC v3 §14 subsection 13.7 for the R7 SFDR PAI axis derivation.

C

Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

ITEM	VALUE
Scoring Engine	SSI Index **v4.2** — Gaussian copula Monte Carlo (10,000 iterations) · 11 modifiers (5 baseline + 6 v4.2 resilience family) · Re composite bounded [0.920, 1.787]
Weight Architecture	C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05
Modifiers Applied	R3 C_mult: 0.944, R4 F_topo: 1.045, R6 restoration: 1.004, R6 seismic: 1.035, R7 cyber: 1.000, R10 just: 1.011, R6d wildfire: 1.030, R6c flood: 1.050, R8 adapt: 0.972, R9 compound: 1.032, R6e winter: 1.010
Degradation Model	Markov 5-state, CIGRE TB 761 calibration
Thermal Model	IEEE C57.91 transformer loading curves
Report Date	17 March 2026
Reporting Period	Calendar year 2025
Refresh Cadence	Annual (data ingestion → scoring engine → display)
Substation	BKW (CH_736626578)
Data Assurance Level	All inputs from institutional open-source with citable vintage
Re composite (v4.2)	Re_raw = 1.1059 · Re_norm = **0.568** · **ELEVATED** exposure · bounds [0.920, 1.787]

D

Re composite — bounded resilience aggregate (v4.2)

Resilience composite anchoring CSRD/ESRS climate-resilience disclosure

The Re composite (v4.2) is a bounded resilience aggregate capturing the new R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just modifier family in a single scalar suitable for CSRD/ESRS climate-resilience disclosure:

```
Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c − 1.00)
```

Bounds: **[0.920, 1.787]**. Re_raw < 1.000 when R8_adapt < 1.00 dominates other near-neutral factors. Swiss fleet observed range across 947 substations: the methodology-bounded extremes [0.920, 1.787] — low-end values arise at interior low-exposure substations with R8_adapt < 1.00 dominating near-neutral other modifiers, high-end values at coastal compound-exposure substations where R6c flood + R9 compound stack multiplicatively. v4.2 explicitly retires the pre-Path-B "winter exposure is a real climate burden" reading per Convention #6 §2.3.1 (Bourgouin freezing-rain p99 tail-event amplifier per §6 design — winter exposure signal correctly lives in v4.0.2 I1 Snow/Ice metric, where it methodologically belongs). See W1-W10 FC v2 for per-substation worked examples + W4 Climate resilience baselines.

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.